Year 13 Further Maths - Complex numbers

Similar questions - Arithmetic, polar form, quadratics

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) In this part you must show detailed reasoning. Solve $4z^2 - 12z + 25 = 0$, giving your answers in modulus-argument form with the argument correct to 3 significant figures. **[5]**

(b) $z_1 = 2 - 5\text{i}$ and $z_2 = 3 + \text{i}$. Express each as exact numbers in the form $a + b\text{i}$: **[5]**

(i) $3z_1 + 2z_2$

(ii) $z_1 z_2$

(iii) $\dfrac{z_1}{z_2}$

(c) Write z_1 in modulus-argument form, giving the modulus exactly and the argument to 3 significant figures. **[3]**

2 In this question you must show detailed reasoning.

(a) Write $-15 + 8\text{i}$ in modulus-argument form, with the argument to 3 significant figures. **[3]**

(b) Solve the simultaneous equations $\text{i}z + 2w = 2 - \text{i}$ and $3z - \text{i}w = 2 + 4\text{i}$, giving z and w in Cartesian form. **[5]**

(c) Determine, in exact form, $\arg\left(3\sqrt{3} + 3\text{i}\right)^4$. **[3]**

Question 1

(a) $z = \dfrac{12 \pm \sqrt{144-400}}{8} = \dfrac{12 \pm 16\mathrm{i}}{8} = \frac{3}{2} \pm 2\mathrm{i}$. Modulus $\sqrt{\frac{9}{4}+4} = \frac{5}{2}$, argument $\arctan\frac{4}{3} = 0.927$. $z = \frac{5}{2}(\cos 0.927 \pm \mathrm{i}\sin 0.927)$

(b) (i) $3z_1 + 2z_2 = 6 - 15\mathrm{i} + 6 + 2\mathrm{i} = 12 - 13\mathrm{i}$. (ii) $z_1 z_2 = 6 + 2\mathrm{i} - 15\mathrm{i} + 5 = 11 - 13\mathrm{i}$. (iii) $\dfrac{(2-5\mathrm{i})(3-\mathrm{i})}{10} = \dfrac{6-2\mathrm{i}-15\mathrm{i}-5}{10} = \dfrac{1}{10} - \dfrac{17}{10}\mathrm{i}$

(c) $|z_1| = \sqrt{29}$, $\arg z_1 = -\arctan\frac{5}{2} = -1.19$. $z_1 = \sqrt{29}(\cos(-1.19) + \mathrm{i}\sin(-1.19))$

Question 2

(a) Modulus $\sqrt{225+64} = 17$. Argument $\pi - \arctan\frac{8}{15} = 2.65$

(b) From the first equation $w = \dfrac{2-\mathrm{i}-\mathrm{i}z}{2}$. Then $\mathrm{i}w = \dfrac{2\mathrm{i}+1+z}{2}$, and the second equation gives $3z - \dfrac{2\mathrm{i}+1+z}{2} = 2+4\mathrm{i}$, so $5z = 5+10\mathrm{i}$, $z = 1+2\mathrm{i}$. Then $w = \dfrac{2-\mathrm{i}-\mathrm{i}+2}{2} = 2-\mathrm{i}$

(c) $\arg(3\sqrt{3}+3\mathrm{i}) = \arctan\dfrac{1}{\sqrt{3}} = \frac{\pi}{6}$, so the argument of the fourth power is $\frac{4\pi}{6} = \frac{2\pi}{3}$